Collaborative Document Day 2 RSST

2026-09-15-research-software-support-training

Welcome to The Workshop Collaborative Document.

This Document is synchronized as you type, so that everyone viewing this page sees the same text. This allows you to collaborate seamlessly on documents.

This is the Document for today: https://edu.nl/jnqpv

Collaborative Document day 1: https://edu.nl/nmwaj

Collaborative Document day 2: https://edu.nl/jnqpv

Collaborative Document day 3: https://edu.nl/g6eqm

Collaborative Document day 4: https://edu.nl/mwc48

Collaborative Document day 5: https://edu.nl/6jm6x

Code of Conduct

Participants are expected to follow these guidelines:

- Use welcoming and inclusive language.
- Be respectful of different viewpoints and experiences.
- Gracefully accept constructive criticism.
- Focus on what is best for the community.
- Show courtesy and respect towards other community members.

If you feel that the code of conduct is breached, please talk to one of the instructors (if the complaint is for one of the participants) or send an email to training@esciencecenter.nl (if the complaint is for one of the instructors).

Certificate of attendance

If you attend the full workshop we will send you a certificate of attendance.
We will delete all personal identifyable information 8 months after the training.

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Getting help

To ask a question, raise your hand in zoom. Click on the icon labeled "Reactions" in the toolbar on the bottom center of your screen,
then click the button 'Raise Hand '. For urgent questions, just unmute and speak up!

You can also ask questions or type 'I need help' in the chat window and helpers will try to help you.
Please note it is not necessary to monitor the chat - the helpers will make sure that relevant questions are addressed in a plenary way.
(By the way, off-topic questions will still be answered in the chat)

Workshop website & Lesson materials

RSST Sept-Oct 2026 (https://esciencecenter-digital-skills.github.io/2026-09-15-ds-rss-masterclass/)

RSST lesson material (https://esciencecenter-digital-skills.github.io/research-software-support/)

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Agenda

Time	Topic
9:00	Welcome and recap
9:30	Software management Plans
10:30	Break
10:45	SMP exercises
11:45	Break
12:00	Version control
13:00	End of Day 2

Homework from last week

- Check if your FAIR mapping was correct: https://www.nature.com/articles/s41597-022-01710-x
- Optional: play around with the coding tool or share with collegues: https://www.online-python.com/
- Optional: further reading materials https://esciencecenter-digital-skills.github.io/research-software-support/modules/softwarelandscape/reading

Collaborative Notes

Software Management Plans (SMPs)

Research producibility:

- Reproductibility relis on re-using research software
- Users need to find, access and use the software
- Users need to understand the software
- Software reusability encompasses both reproducing existing results as well as building upon existing software to use it in new or different ways
- SMP is a guideline to help people think how to achieve this

SMP

- Build on the success of a Data Management Plans (DMPs)
- A document detailing how research software will be managed
    - What does it do?
    - Who is it for?
    - What resources does it need?
    - Who is responsible?
    - How long will it be availabke?
- Can be part of a project proposal, or generated in the early phases. Funders have started to adopt it as a requirement.
- It is a "living document" - updated as plans change!

An SMP helps...

- Helps researchers explore existing software that could fit the purpose (with some tweaking)
- It helps plan resources (time/money) - helps make it a recosngiseable job; researcher might work a lot on software development, but it's not always seen as labour. An SMP can highlight it is a product of the research project.
- It avoids issues later to make software FAIR backwards
- Preparing software for modular integration
- It makes technical choices explicit
- SMP helps keep research software alive

An SMP is not...

- Another bureaucratic hoop for researchers to jump through.
- Created an online tool/decision tree to make it easier for people to interact

SMP becomes more rigorous as the software becomes more complex

A practical guide to SMP's: https://doi.org/10.5281/zenodo.7038280

SMP Guided Questionnaire: https://smp.research.software/

- Determine scope (4 questions)
- Fill in SMP
- Download a word document or machine readable format

Which SMP To Use?

- Check your institution - do you already have a template? (ask your local Digital

Competence Center)
- It might also include an assessment rubric with (un)acceptable answers (like; in a medical context the rigour required of software might change)

Version control

You might now version control from document history.
You can save your changes in your document of code in commits.
Each commit contains a commit message, and it has some metadata/description of the commite and the version of the software attached to it.
Pick a relevant commit size for your project (1 typo might be too small for a new commit, multiple new features too big, those can better be split in multiple sepearate commits).
A series of commits form a log, which can be saved in a repository.
The 2 most often used version schemes are:

- semantic versioning
- calendar versioning

Each of these options has some pro's and con's.
Sometimes the community uses another scheme for versioning.

Questions from this week

- For some departments it makes more sense to do an SMP than a DMP, but we hear funders still require DMP and they should name their SMP as such to meet that requirement
    - A DMP indeed remains formal requirement, but funders are lobbying and exploring
- (Miriam Kos) At Radboud University we are in the phase of launching a SMP template soon (Digital Competence Centre in collaboration with my centre Donders Centre for Cognition). End of this year we will evaluate and start building the mix-template of DMP/SMP. If others have already done this, or are also working on this, would be great to hear about this.
- (breakout room 5, first assignment) when reusing code (a chunk of it), how do you reference to it?

Exercises

What's in it for me? Convince your colleague on why an SMP is important (10 minutes)

https://esciencecenter-digital-skills.github.io/research-software-support/modules/softwaremanagementplans/smp_importance

Room 1

software: NEMO

- against
  Researchers were busy, no time to make a SMP
  Technical Knowledge Barrie, What exaclty is a SMP? Where to start?

Lack of direct incentives

- for
  Reuse of the software
  Low barrier of adoption can help implementation

Room 2

- against
    - why DMP and SMP are two separate documents, why there is no structured and standartised temlate, flexible to adjust to the research projects in one go
    - why do we need SMP if it is not required by the funders, and if I am publishing the code in the repo
- for
    - SMP is guidance, not obligation
    - small projects don't need a big SMP. SMP adjusts to the size of the project
    - Depending on the type of research SMP is sometimes more relevant (for researchers mainly developing software).
    - It can help them foresee their needs for their course of project
    - A DMP not detailed enough for software-based research.

Room 3

- software: measure average height of students in class
- for:
    - What about reusing it next year
    - Can use the plan to structure the methods of the paper
    - Useful for reproducability
    - Use the jupyter notebook itself as a (part of the) "smp" in a way
    - Ask the person who wrote the code (student) to write the smp
    - Even if you're not interested in reusing the code later, another researcher might
    - With a proper smp you are probably more likely to be cited
- against:
    - Simply one notebook/script
    - No time for this
    - Paper is already written and never asked about this smp
    - Unlikely for "our" research group to return to this research
    - It was my student and not me who made the code
    - Feels like its an investment on my part as a researchre but more of a benefit to others, i.e., not worth my time

Room 4

- For:
    - Programs to do statistics? Already software!

    - R scripts from work could be reused by colleagues
    - SMP sounds like a bureaucratic hurdle, but it's like regular house-cleaning - doing a bit now helps you in the long run
    - Benefit of SMP is that it's about writing the software and making adjustment, how to adjust mistakes, how to do version control; but also about sharing software in the future to make it easier for others to reuse in the future
    - SMP doesn't have to be a big complex form, can just be a thought process
    - Even for more simple scripts, you don't have to go crazy with detail or efforts with remote repo's; just focus on the SMP as a way to help you and as a way to publish your software later for it to be available alongside your research
    - Focus on the SMP as a way to build on your work, it could be mean more citations of your work

- Against:

    - Too busy
    - Just doing healthcare, not doing software!
    - My scripts are just standard, what everybody else uses!
    - Already documented what I'm doing with software in the DMP and in the protocol!
    - Simple analysis scripts don't feel like they warrant SMPs and thinking of this stuff in detail

Room 5

- first explanation of SMP, difference with DMP
  -how SMP is stored, how it is used
  -show SMP template
  -when reusing code (a chunk of it), how do you reference to it?
- For:
  -publishers ask for code
  -researchers will continue your project after you, and SMP will help think you how they can understand and reuse your code
  -SMP asks questions you might have not thought about it yet, it helps you plan project, learn about good practices, think about managing your code
  -code is also regarded as research output. Sharing it can helps assess the quality of research
  -The more details you give, reviewers really appreciate this. Sharing code speaks for you as a researcher, that you are open.
  -at some point everyone is asked to share code
  -publishing code gets you cited in other papers.
- Against:
  -Cannot maintain
  -Don't know who will be using it
  -Isn't this already covered in DMP
  -my supervisors are not convinced of use of SMP, it is a lot of effort. They don't know coding, they don't have experience in this.
  -what if my code is not very high quality, wouldn't is be a disadvantage to share it? More room for criticism.

Exploring the SMP Questionnaire

https://esciencecenter-digital-skills.github.io/research-software-support/modules/softwaremanagementplans/smp_questionnaire

Room 1

Room 2

Room 3

Room 4

Room 5

Note: Write a SMP for the whole PhD or for each paper?

Reuse vs Users?

SMP vs FAIR

https://esciencecenter-digital-skills.github.io/research-software-support/modules/softwaremanagementplans/smpvsfair

Room 1

Accuracy of results - SMP or FAIR not clear
Reproducibility of results - FAIR and SMP together
Reusability of software - FAIR and SMP
User friendliness of software - Hard to choose, not covering.
Software quality - SMP
Recognition and rewards for software developers - SMP
Scholarly archive - FAIR and SMP
Software sustainability - FAIR and SMP

Room 2

Accuracy of results - score:1 ; SMP
Reproducibility of results - score:1 ; FAIR&SMP
Reusability of software - score:2; FAIR&SMP
User friendliness of software - score: 3; SMP (user documentation)
Software quality - score: 1; SMP
Recognition and rewards for software developers-score:3 ; FAIR&SMP
Scholarly archive - score:3 ; FAIR
Software sustainability -score: SMP

Room 3

Verification of results - 1
Accuracy of results - score:1 ; SMP
Reproducibility of results - score:1 ; FAIR&SMP
Reusability of software - score:1; FAIR&SMP
User friendliness of software - score: 1; SMP (user documentation)
Software quality - score: 1; SMP

Recognition and rewards for software developers-score:2 ; SMP
Scholarly archive - score:3 ; FAIR
Software sustainability -score: SMP

Room 4

Motivations

- Arguments could be made that almost everything has overlap
    - lots of things to make software more FAIR can impact quality/accuracy etc although FAIR itself doesn't directly impact quality
- Overlapping:
    - Reproducibility
    - Reusability
    - Recognition and Rewards (via citations via findability)
    - Scholarly archive
    - *maybe* user friendliness as part of reusability (*some discussion here*)
- FAIR principles-only:
    - None of them?
- SMPs only:
    -

Room 5

What to include in the repo?

https://esciencecenter-digital-skills.github.io/research-software-support/modules/version-control/version_control_exercise

Discuss in your break-out room which of the following should be part of the code repository and why it should or should not.

- installation instructions (i.e. detailed instructions on how to install the software)
- user documentation (i.e. information for the users of the software about its functionality and features)
- full dataset to reproduce figures from a paper
- a few data files to test the functionality of the code base
- in-line comments intended only for code developers

Room 2

installation instructions (i.e. detailed instructions on how to install the software) :heavy_check_mark:
user documentation (i.e. information for the users of the software about its functionality and features) :heavy_check_mark:
full dataset to reproduce figures from a paper :x:
a few data files to test the functionality of the code base :heavy_check_mark:
in-line comments intended only for code developers :x: (can cause confusion)

Resources

- Overview of landscape: https://lcrdm.nl/wp-content/uploads/2023/12/RDMRSE-in-NL-32-1.pdf
- SMP Decision Tree
    - Online questionnaire tool: https://smp.research.software
    - Source files: https://github.com/SS-NES/docassemble-SMPDecisionTree (Apache 2.0 licensed)
        - The specific files with the questions and decision help explanatory texts can be found in the docassemble/SMPDecisionTree/data/questions (https://github.com/SS-NES/docassemble-SMPDecisionTree/tree/main/docassemble/SMPDecisionTree/data/questior folder
- Versioning Schemes
    - Semantic Versioning (SemVer): https://semver.org
    - Calendar Versioning (CalVer): https://calver.org
- Discussion slides on Codeberg as alternative to GitHub: https://zenodo.org/records/22877131

Parking lot

Homework

- If you have time, please finish your SMP. https://esciencecenter-digital-skills.github.io/research-software-support/modules/softwaremanagementplans/smp_questionnaire. No need to hand them in (we are not checking you), but we suggest you go over the remaining points. We reserve a bit of time at the start of day 3 to go over any questions about SMPs.
- (optional) Check the resources mentioned in this collab doc and from the lesson material here: https://esciencecenter-digital-skills.github.io/research-software-support/modules/softwaremanagementplans/reading
- (optional) Check the resources related to version control, including the git commit cheat sheet: https://esciencecenter-digital-skills.github.io/research-software-support/modules/version-control/version_control_resources